

Exercise (From HOTS): I will roll a single die no more than three times. You can stop me immediately after the first roll, or immediately after the second, or you can wait for the third. I will pay you the same number of dollars as there are dots on the single upturned face on my last roll (roll number three unless you stop me sooner). What is your playing strategy?

Consider the " (a, b) strategy": Stop after roll 1 if it's at least a , stop after roll 2 if it's at least b .

Let X = winnings

$A_i = \{\text{stop after toss } i\} \quad i=1, 2, 3$

$Y_i = \text{outcome of toss } i$

Fix a, b such that $a, b \in \{1, 2, \dots, 6\}$

$$E(X) = E(X|A_1)P(A_1) + E(X|A_2)P(A_2) + E(X|A_3)P(A_3)$$

$$E(X|A_1) = E(Y_1 | Y_1 \geq a) = \sum_{k=1}^6 k P(Y_1 = k | Y_1 \geq a)$$

$$= \sum_{k=1}^6 k \frac{P(Y_1 = k, Y_1 \geq a)}{P(Y_1 \geq a)}$$

$$= \sum_{k=a}^6 k \frac{\frac{1}{6}}{\left(\frac{7-a}{6}\right)} = \left(\frac{1}{7-a}\right) \left(21 - \frac{a(a-1)}{2}\right)$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$E(X|A_2) = \left(\frac{1}{7-b}\right) \left(21 - \frac{b(b-1)}{2}\right)$$

$$E(X|A_3) = 3.5$$

$$P(A_1) = P(Y_1 \geq a) = \frac{7-a}{6}$$

$$P(A_2) = P(Y_1 < a, Y_2 \geq b) = \left(\frac{a-1}{6}\right) \left(\frac{7-b}{6}\right)$$

Since tosses are independent

$$P(A_3) = P(Y_1 < a, Y_2 < b) = \left(\frac{a-1}{6}\right) \left(\frac{b-1}{6}\right)$$

Optimal is $a=5, b=4$

b

	[, 1]	[, 2]	[, 3]	[, 4]	[, 5]	[, 6]
[1,]	3.500000	3.500000	3.500000	3.500000	3.500000	3.500000
[2,]	3.916667	3.986111	4.027778	4.041667	4.027778	3.986111
[3,]	4.166667	4.305556	4.388889	4.416667	4.388889	4.305556
[4,]	4.250000	4.458333	4.583333	4.625000	4.583333	4.458333
[5,]	4.166667	4.444444	4.611111	4.666667	4.611111	4.444444
[6,]	3.916667	4.263889	4.472222	4.541667	4.472222	4.263889

Exercise (From HOTS): A coin-making machine produces pennies. Each penny is manufactured to have a probability P of turning up heads. However, the machine draws P randomly from the uniform distribution on $[0, 1]$ so P can differ for each coin produced. A coin pops out of the machine. You flip it once, and it comes up heads. Given this information, what is the (conditional) cumulative distribution function $F_{P|H}(p)$ for the probability of a head for that coin (where "H" denotes conditioning on the head)?

What is the (conditional) distribution for the probability of a head if you flip the coin 1,000 times and get 750 heads?

Assume make n flips. Let $X = \#$ of heads, $0 \leq P \leq 1$
Find $P(P \leq p | X = x)$. conditional CDF

$$\text{Note that } P(P \leq p | X = x) = \int_0^p f_{P|X}(u|x) du$$

$$= \int_0^p \left[\frac{f_{X|P}(x|u) f_P(u)}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv} \right] du$$

$$\text{since, } f_{P|X} = \frac{f_{P,X}}{f_X} = \frac{f_{X|P} f_P}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv}$$

know $f_P(u) = 1$, $0 \leq u \leq 1$

$$f_{X|P}(x|u) = \binom{n}{x} u^x (1-u)^{n-x}, \quad x=0,1,\dots,n$$

$$S_0 = \frac{\int_0^1 \binom{n}{x} u^x (1-u)^{n-x} (1) du}{\int_0^1 \binom{n}{x} v^x (1-v)^{n-x} (1) dv}$$

$$= \int_0^1 \underbrace{\left(\frac{\Gamma(n+1)}{\Gamma(x+1) \Gamma(n-x+1)} \right)}_{f_{P|X}} u^x (1-u)^{n-x} du$$

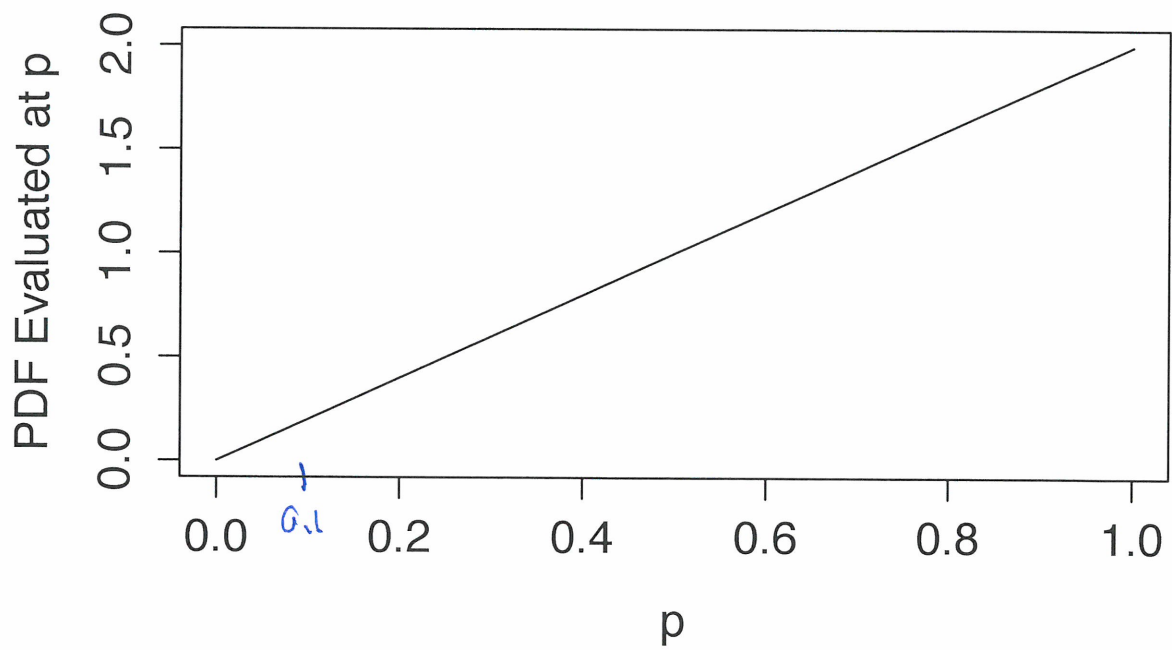
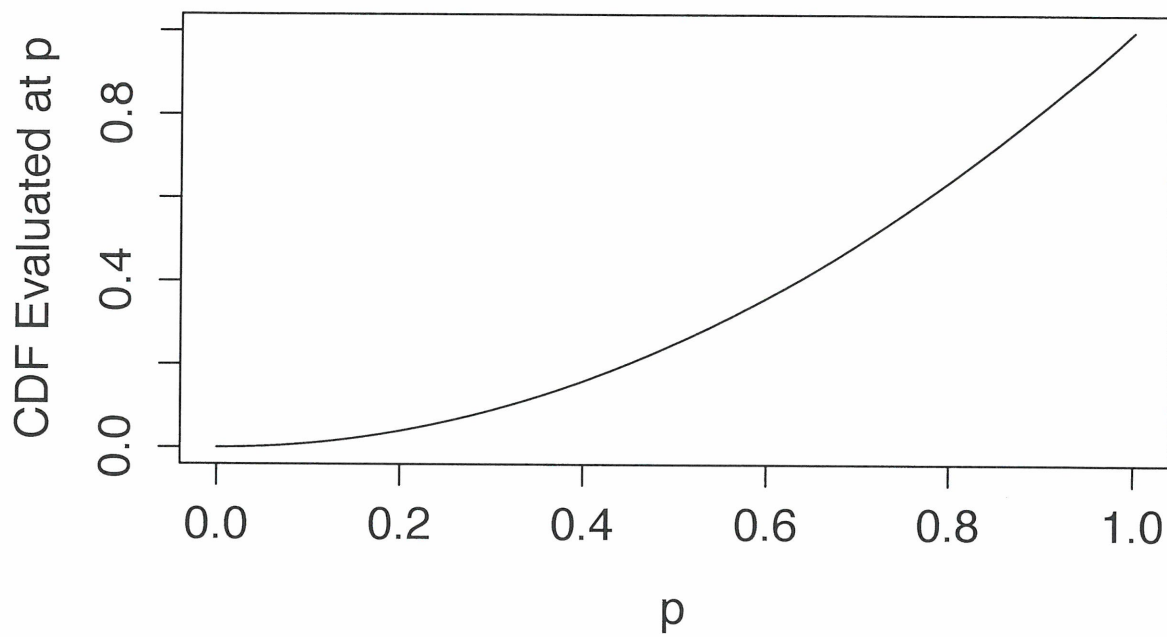
Math Facts

$$= P(U \leq p) \quad U \sim \text{Beta}(x+1, n-x+1)$$

Given x heads in n flips, P has
Beta($x+1$, $n-x+1$) distribution

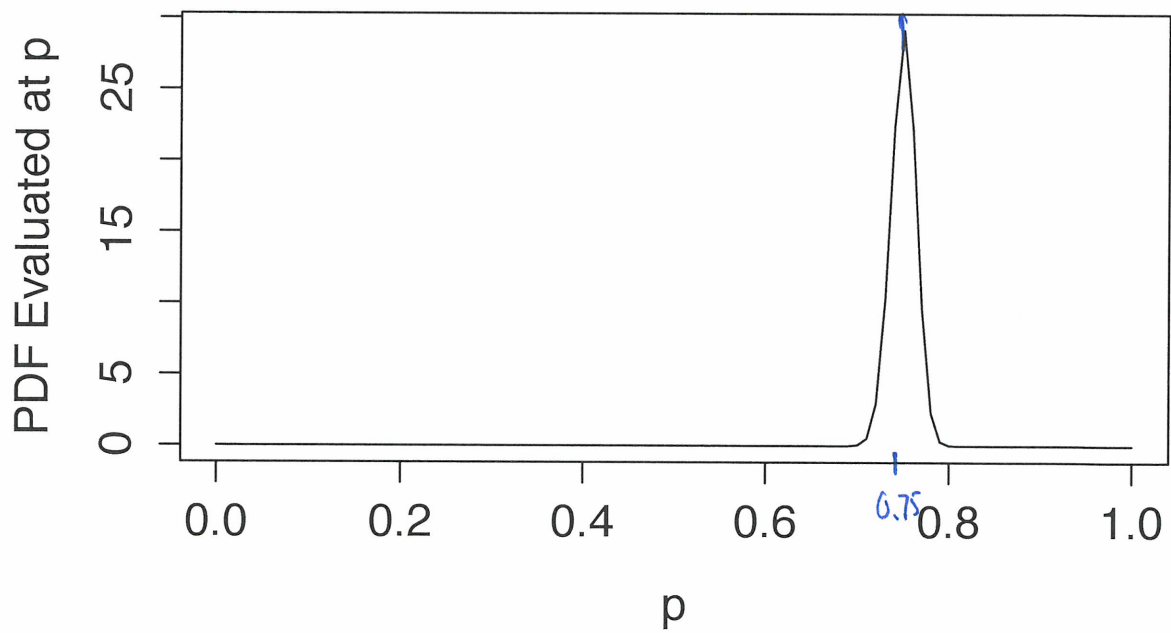
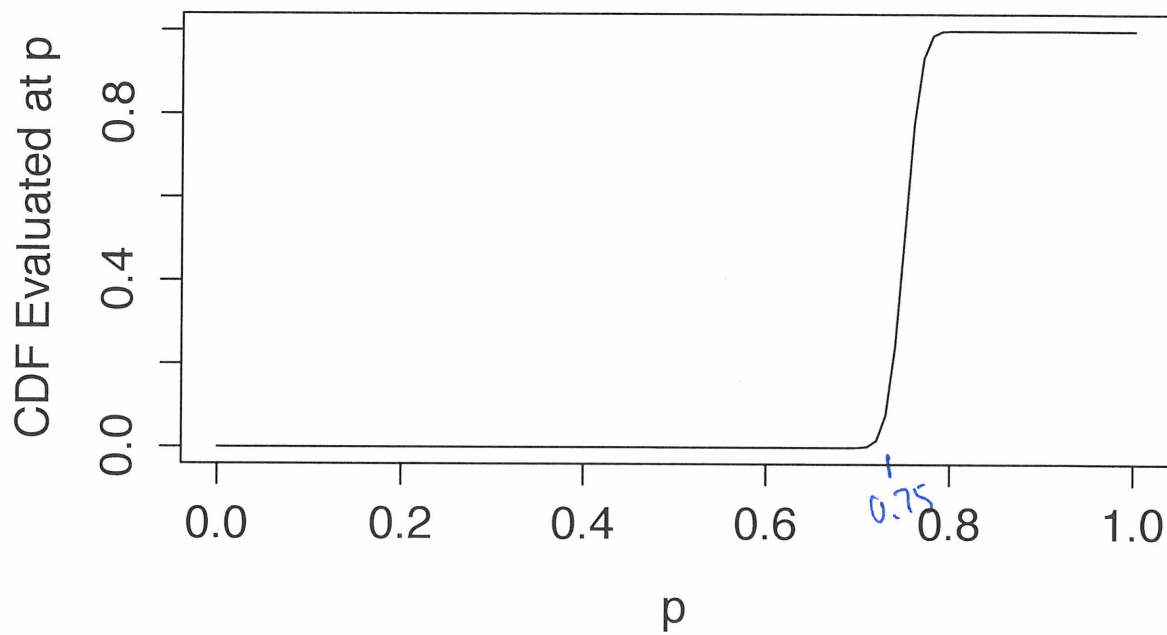
If $n=1$, $x=1$, $P \sim \text{Beta}(2, 1)$
CDF is p^2 on $0 \leq p \leq 1$

1 For the case where $n = 1$ and $x = 1$:



2

1 For the case where $n = 1000$ and $x = 750$:



2

Iterated Conditioning

There is convenient shorthand for the above version of the Law for Total Probability for Expectations. We often write

$$E(X) = E(E(X|Y))$$

This expression works for both discrete and continuous random variables.

compare to $E(X) = \int_{-\infty}^{\infty} E(X|Y=y) f_Y(y) dy$

It also holds that

$$V(X) = V(E(X|Y)) + E(V(X|Y))$$

Exercise: Explain the logic behind the iterated conditioning formula.

Note that $E(X|Y)$ is a RV, a function on Ω

Let $Z = E(X|Y)$, $Z(\omega) = E(X | Y = Y(\omega))$

i.e. Z captures all the possible values of Y that we could condition on for outcome $\omega \in \Omega$, $Y = Y(\omega)$

So, $E(Z)$ takes a weighted average, of all possible ω (all possible $Y(\omega)$) with weighting provided by f_Y

Exercise: The lifetime of a part can be modelled using an Exponential(λ) distribution. These parts are produced in lots, and it is most realistic to assume that λ varies from lot to lot. Based on experience, the manufacturer models the lot-to-lot variability in λ using the pdf

$$f_{\Lambda}(\lambda) = 4\lambda e^{-2\lambda}, \quad \lambda > 0 \quad \Lambda \sim \text{Gamma}(2, 2)$$

Determine the unconditional probability density function for the length of life of a part. What is the probability that a randomly chosen part lasts longer than 2 time units? What is the expected length of the life of a part?

$X = \text{length of life of randomly chosen part}$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|\Lambda}(x|\lambda) f_{\Lambda}(\lambda) d\lambda$$

$$= \int_0^{\infty} (\lambda e^{-\lambda x}) (4\lambda e^{-2\lambda}) d\lambda, \quad x > 0$$

$$= \frac{8}{(x+2)^3}, \quad x > 0 \quad (\text{see Math Facts})$$

$$P(X > 2) = \int_2^{\infty} \frac{8}{(x+2)^3} dx$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \dots$$

Or: $E(X) = E(E(X | \Lambda))$

$$= E\left(\frac{1}{\Lambda}\right)$$

$$= \int_0^{\infty} \frac{1}{\lambda} 4\lambda e^{-2\lambda} d\lambda$$

$$= 2 \underbrace{\int_0^{\infty} 2e^{-2\lambda} d\lambda}_{=1}$$

← exponential(2) density

$$= 2$$

Exercise: Consider a Poisson process with rate λ . Suppose that “events” are of two types: “Type I” and “Type II.” An event is of Type I with probability p , and the type is determined independently from event to event. Let X denote the number of events in one time unit, so that X is $\text{Poisson}(\lambda)$. Let Y denote the number of events in one time unit that are of Type I. Find $E(Y)$ and $V(Y)$.

$$E(Y) = E(E(Y|X))$$

$$= E(Xp) \quad Y|X \sim \text{binomial}(X, p)$$

$$= pE(X) = p\lambda, \text{ since } X \sim \text{Poisson}(\lambda)$$

$$V(Y) = V(E(Y|X)) + E(V(Y|X))$$

$$= V(Xp) + E(Xp(1-p))$$

$$= p^2 V(X) + p(1-p) E(X)$$

$$= p^2 \lambda + p(1-p) \lambda$$

$$= \lambda p$$

Is $Y \sim \text{Poisson}(\lambda p)$? Yes, but we have not proven that yet.

$$V(Y) = E(Y^2) - E(Y)^2$$

$$= E(E(Y^2|X)) - E(E(Y|X))^2$$

$$= E(V(Y|X) + E(Y|X)^2) - E(E(Y|X))^2$$

$$E(Z^2) = V(Z) + E(Z)^2$$

$$= E(V(Y|X)) + E(E(Y|X)^2) - E(E(Y|X))^2$$

$$= E(V(Y|X)) + V(E(Y|X))$$

1 **Exercise (Interview Question):** Suppose that the random variables A , B ,
2 and C are i.i.d. $\text{Uniform}(0, 1)$. What is the probability that

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$$Ax^2 + 2Bx + C = 0$$

4 has roots which are both real?

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Measure-Theoretic Notions of Conditional Expectation

The statement that we will “condition on Y ,” implies that one should behave as if the value of Y is “known.”

We need to formalize this notion beyond conditioning on a single random variable, as this is crucial to more advanced study of probability. We will consider conditioning on more general forms of “information.”

It is useful to consider the following question: Given the nature of the information to be provided, for which subsets of the sample space Ω am I certain of their occurrence/non-occurrence?

The following example will help to build some intuition.

- 1 **Exercise:** Suppose that I will toss a coin three times. Let X denote the
 2 number of "heads." If I tell you that the information I will provide is the
 3 value of X , for which subsets of Ω are you certain of their occurrence/non-
 4 occurrence, regardless of the value of X ?

ω	$X(\omega)$	
HHH	3	$A_1 = \{HHH\}$
HHT	2	$A_2 = \{HHT, HTH, THH\}$
HTH	2	$A_3 = \{HTT, THT, TTH\}$
THH	2	$A_4 = \{TTT\}$
HTT	1	
THT	1	For each of A_1, A_2, A_3, A_4
TTH	1	know occ/non-occ for
TTT	0	any value of X

14
 15 But, will also know occ/non-occ of
 16 $\emptyset, A_1 \cup A_2, A_1 \cup A_3, \dots, A_1 \cup A_2 \cup A_3, \dots, \Omega$

17
 18 would not know occ/non-occ status
 19 of $B = \{HHT\}$. If info. is $X=2$,
 20 then not sure if B has occurred or not.

21
 22 Denote this collection $\mathcal{F}$

1 The above example demonstrates how “information” is captured via a col-
2 lection of subsets of Ω ; we will generally denote such collections using $\mathcal{F}$,
3 $\mathcal{G}$ or similar.

4 We will write $E(X|\mathcal{F})$ to indicate that we are conditioning on all of the
5 information in the set $\mathcal{F}$.

6 A typical application of this idea is when

7
$$\mathcal{F}_t = \text{information available at time } t$$

8 Then, we have that

9
$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

10 This sequence $\{\mathcal{F}_t\}$ is called a **filtration**.

11 For example, any trading strategy implemented at time t should condition
12 on $\mathcal{F}_t$, but not on $\mathcal{F}_{t+1}$.

Formally, such collections $\mathcal{F}$ are a σ -field or σ -algebra. A σ -**field** is a collection of subsets of Ω that satisfies three properties:

1. $\emptyset \in \mathcal{F}$
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
3. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$

Exercise: Comment on how each of the three properties of a σ -field relate to the interpretation that $\mathcal{F}$ represents information.

① I always know $\emptyset$ has not occurred

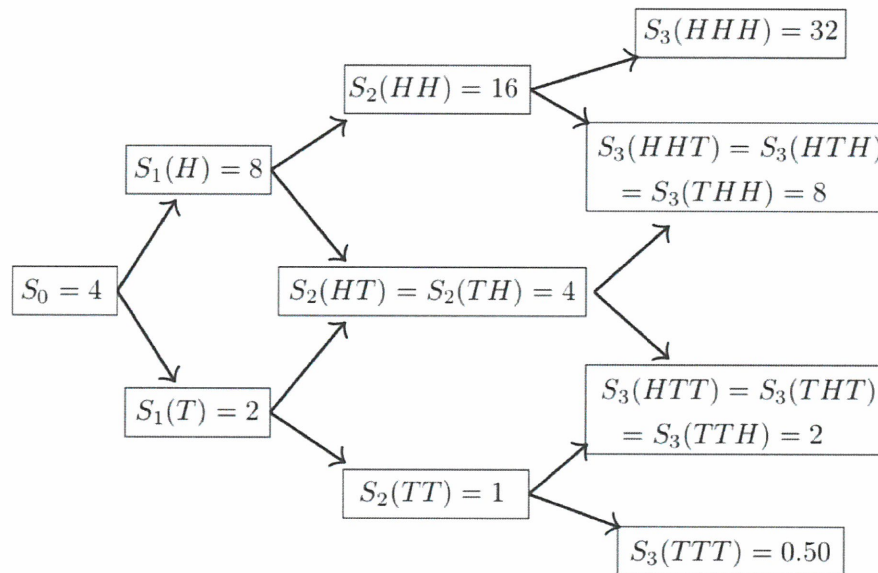
② If I know occ/non-occ of A ,
then also know occ/non-occ of A^c

③ If I know occ/non-occ of $A_1, A_2, \dots$
then I know occ/non-occ of $\bigcup_{i=1}^{\infty} A_i$

Comments

1. Be careful with the notation and the interpretation: If $\mathcal{F}$ is a σ -field, its elements are subsets of Ω , i.e., if $A \in \mathcal{F}$, then $A \subseteq \Omega$.
2. The **trivial σ -field** consists of only $\emptyset$ and Ω . So, if $\mathcal{F}_0 = \{\emptyset, \Omega\}$, then conditioning on $\mathcal{F}_0$ is like conditioning on no information.
3. The **power set**, denoted 2^Ω , is a σ -field. It corresponds to “knowing everything.”
4. The information content from conditioning on a random variable Y is called the **σ -field generated by Y** , and is denoted $\sigma(Y)$. So, in our initial example, we were listing the sets in $\sigma(X)$. When we write $E(Y|X)$, that can be interpreted as $E(Y|\sigma(X))$.
5. A random variable X is said to be **$\mathcal{F}$ -measurable** if $\sigma(X) \subseteq \mathcal{F}$. Heuristically, this means that the information on which I am conditioning includes the value of X .

- 1 **Exercise:** Consider a simple three-period binomial pricing model. At each
 2 step, the price doubles with probability 0.5, and is cut in half with proba-
 3 bility 0.5. All steps are independent.
- 4 Start by constructing an appropriate Ω for this situation.
- 5 Let S_t denote the value at time t , and assume that $S_0 = 4$.
- 6 Find $E(S_t)$ for all values of t . What is $E(S_2|\mathcal{F}_1)$? What is $E(S_3|\mathcal{F}_1)$?



$$\Omega = \{HHH, HHT, \dots, TTT\}$$

notation $\omega \in \Omega$ write $\omega = \omega_1 \omega_2 \omega_3$
 $\omega_i = H \text{ or } T$

$$S_0(\omega) = 4 \quad \text{for all } \omega \in \Omega$$

$$S_1(\omega) = \begin{cases} 8, & \omega_1 = H \\ 2, & \omega_1 = T \end{cases}$$

$$E(S_0) = 4, \quad E(S_1) = 5, \quad E(S_2) = \frac{25}{4},$$

What is $E(S_2 | \mathcal{F}_1)$? This is a RV.
It is a function on Ω .

If I know $\omega_1 = H$, then 50% chance
 $S_2 = 16$ and 50% chance that $S_2 = 4$.

If I know $\omega_1 = T$, then 50% chance
 $S_2 = 4$ and 50% chance that $S_2 = 1$

When $\omega_1 = H$, EV for $S_2 = (16)(\frac{1}{2}) + (4)(\frac{1}{2}) = 10$

When $\omega_1 = T$, EV = $\frac{5}{2}$

So,

$$E(S_2 | \mathcal{F}_1) = \begin{cases} 10, & \omega \in \{HHH, HHT, \\ & HTH, HTT\} \\ 5/2, & \omega \in \{THH, TTH, \\ & THT, TTT\} \end{cases}$$

Note that $\mathcal{F}_1$ contains $\emptyset, \Omega,$
 $\{HHH, HHT, HTH, HTT\}, \{THH, TTH, THT, TTT\}$
 and that's it

Here, $\sigma(S_1) = \mathcal{F}_1$

Does it hold that

$$E\left(\underbrace{E(S_2 | \mathcal{F}_1)}\right) = E(S_2) ?$$

Yes.

$$(10)\left(\frac{1}{2}\right) + \left(\frac{5}{2}\right)\left(\frac{1}{2}\right) = \frac{25}{4} = E(S_2)$$

$$E(S_3 | \mathcal{F}_1)(\omega) = (32)\left(\frac{1}{4}\right) + (8)\left(\frac{1}{2}\right) + (2)\left(\frac{1}{4}\right) \\ = 12.5$$

when $\omega_1 = H$, i.e. $\omega \in \{HHH, HHT, HTH, HTT\}$

$$E(S_3 | \mathcal{F}_1)(\omega) = (8)\left(\frac{1}{4}\right) + (2)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) \\ = \frac{25}{8} \quad \text{when } \omega_1 = T$$

$E(S_3 | \mathcal{F}_1)$ is a RV

$$E(S_3) = E(E(S_3 | \mathcal{F}_1))$$